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Pegs and Disks Problem (3 Pegs, 3 Disks)

Three pegs A, B and C and three disks of different sizes are given. Disk 1 is the smallest and disk 3 is the largest. Initially all the disks are on peg A, with the largest at the bottom. The goal is to bring all the disks to peg B. Only the topmost disk of a peg can be moved, and it can be placed on top of any other peg. As in the standard puzzle, a larger disk is never placed on a smaller disk.

1. Problem Formulation

State representation: A state is written as (position of disk 3, position of disk 2, position of disk 1), where each position is A, B or C. Since a larger disk can never be on a smaller one, this tuple gives the exact arrangement of every peg.

Initial state: (A, A, A) – all disks on peg A (disk 3 at the bottom, disk 1 on top).

Goal state: (B, B, B) – all disks on peg B in the same order.

Operators: Move the topmost disk of any peg to the top of any other peg (A to B, A to C, B to A, B to C, C to A, C to B), provided it is not placed on a smaller disk.

Path cost: Each move costs 1, so the cost of a solution is the number of moves.

State space: 3 pegs and 3 disks give $3^3 = 27$ possible states.

2. Solution (Sequence of Moves)

(Disks on each peg are listed from bottom to top.)

Initial state

A: 3 2 1 | B: empty | C: empty State: (A, A, A)

Move 1: disk 1 from A to B

A: 3 2 | B: 1 | C: empty State: (A, A, B)

Move 2: disk 2 from A to C

A: 3 | B: 1 | C: 2 State: (A, C, B)

Move 3: disk 1 from B to C

A: 3 | B: empty | C: 2 1 State: (A, C, C)

Move 4: disk 3 from A to B

A: empty | B: 3 | C: 2 1 State: (B, C, C)

Move 5: disk 1 from C to A

A: 1 | B: 3 | C: 2 State: (B, C, A)

Move 6: disk 2 from C to B

A: 1 | B: 3 2 | C: empty State: (B, B, A)

Move 7: disk 1 from A to B

A: empty | B: 3 2 1 | C: empty State: (B, B, B) – Goal reached

3. Path in the State Space

(A,A,A) – (A,A,B) – (A,C,B) – (A,C,C) – (B,C,C) – (B,C,A) – (B,B,A) – (B,B,B)

4. Result

The goal state is reached in **7 moves**, which is $2^3 - 1$, the minimum possible for 3 disks. From the initial state only two moves are legal (disk 1 to B or disk 1 to C), and a breadth-first search over the 27 states also gives this 7-move path, so it is the optimal solution.